Lecture 3. Cholesky Decomposition

Existence and uniqueness, algorithm and cost, pivoting, applications, sweep operator

Contents

3.1 The Cholesky decomposition **2**

3.2 Applications **3**

3.3 The sweep operator **3**

3.4 Selected exercises **4**

Mantra 2

The structure should be exploited whenever possible in solving a problem.

Common structures: symmetry, positive (semi)definiteness, sparsity, Kronecker products, low rank. Statisticians mostly meet positive (semi)definite matrices (covariances, Gram matrices $X^\top X$): $x^\top Mx > 0$ (≥ 0) for $x \neq 0$, and since $x^\top Mx = x^\top \frac{M+M^\top}{2} x$ we may assume M symmetric.

3.1 The Cholesky decomposition

Motivation. Cholesky factors a symmetric positive definite A as $LL^\top$ at half the cost of LU, stably and without pivoting.

Theorem 3.1 (Cholesky)

If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, then $A = LL^\top$ for a unique lower triangular L with positive diagonal entries.

Proof. Induction on n. For $n = 1$, $a = \sqrt{a}\sqrt{a}$ with $a > 0$. For $n > 1$, partition and match blocks:

$$\begin{pmatrix} a_{11} & \mathbf{a}^\top \\ \mathbf{a} & A_{22} \end{pmatrix} = \begin{pmatrix} \ell_{11} & \mathbf{0}^\top \\ \mathbf{b} & L_{22} \end{pmatrix} \begin{pmatrix} \ell_{11} & \mathbf{b}^\top \\ \mathbf{0} & L_{22}^\top \end{pmatrix} \iff a_{11} = \ell_{11}^2, \quad \mathbf{a} = \ell_{11}\mathbf{b}, \quad A_{22} = \mathbf{b}\mathbf{b}^\top + L_{22}L_{22}^\top.$$

As $a_{11} = e_1^\top Ae_1 > 0$, $\ell_{11} = \sqrt{a_{11}}$ and $\mathbf{b} = \mathbf{a}/\sqrt{a_{11}}$ are forced. The **Schur complement** $S = A_{22} - \mathbf{b}\mathbf{b}^\top = A_{22} - \mathbf{a}\mathbf{a}^\top/a_{11}$ is positive definite: for $\mathbf{y} \neq \mathbf{0}$ put $x = (-\mathbf{a}^\top\mathbf{y}/a_{11}, \mathbf{y}^\top)^\top \neq 0$; then

$$0 < x^\top Ax = \frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} - 2\frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} + \mathbf{y}^\top A_{22}\mathbf{y} = \mathbf{y}^\top S\mathbf{y}.$$

By induction $S = L_{22}L_{22}^\top$ for a unique such L_{22}. ■

The proof is the algorithm: first column of L, Schur complement update, recurse, in place in the lower triangle of A.

Cost. By symmetry only half the update entries are computed:

$$\tfrac{1}{2}\{2(n-1)^2 + 2(n-2)^2 + \cdots + 2\} \approx \tfrac{1}{3}n^3 \text{ flops}$$

plus n square roots: *half* the cost of LU. It is very stable without pivoting and fails (a nonpositive number under a root) exactly when A is not positive definite, so it is the cheapest *test* of positive definiteness.

Example 3.1

For

$$A = \begin{pmatrix} 4 & 6 & 4 & 4 \\ 6 & 37/4 & 11/2 & 6 \\ 4 & 11/2 & 29/4 & 7 \\ 4 & 6 & 7 & 33/4 \end{pmatrix}$$

the first column of L is $(4,6,4,4)^\top/2 = (2,3,2,2)^\top$. The Schur complement has first column $(37/4 - 9,\ 11/2 - 6,\ 6 - 6)^\top = (1/4, -1/2, 0)^\top$, so $\ell_{22} = 1/2$ and the second column of L is

$(1/2, -1, 0)^\top$. The pivots are $4, \frac{1}{4}, \frac{9}{4}, \frac{1}{4}$ and

$$L = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 3 & 1/2 & 0 & 0 \\ 2 & -1 & 3/2 & 0 \\ 2 & 0 & 2 & 1/2 \end{pmatrix}.$$

All pivots are positive, so A is positive definite, and $\det A = (2 \cdot \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{2})^2 = \frac{9}{16}$.

Pivoting for semidefinite matrices. If A is only positive semidefinite (e.g. $X^\top X$, X rank deficient), a zero pivot can occur. **Symmetric pivoting** moves the largest remaining diagonal entry to the pivot position by a symmetric row and column swap. If it is 0, the remaining block is 0 (since $|a_{ij}| \le \sqrt{a_{ii} a_{jj}}$) and the algorithm stops:

$$PAP^\top = LL^\top, \qquad L \in \mathbb{R}^{n \times r}, \quad r = \operatorname{rank} A.$$

LAPACK provides `potrf` (no pivoting) and `pstrf` (pivoting); R's `chol` returns the upper factor $R = L^\top$, $A = R^\top R$.

3.2 Applications

Multivariate normal log-likelihood. For $y \sim N_n(0, \Sigma)$,

$$\log f(y) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log\det \Sigma - \frac{1}{2} y^\top \Sigma^{-1} y.$$

- Method 1: invert Σ ($2n^3$), form the quadratic form ($2n^2$), compute the determinant by LU ($\frac{2}{3}n^3$).
- Method 2: factor $\Sigma = LL^\top$ ($\frac{1}{3}n^3$), solve $Lx = y$ by forward substitution (n^2), then $y^\top \Sigma^{-1} y = x^\top x$ ($2n$) and $\log\det \Sigma = 2 \sum_i \log \ell_{ii}$ (n).

Method 2 is about 8 times cheaper, more accurate, and its log-determinant cannot overflow. For many evaluation points, factor once and pay n^2 per point.

Linear regression. For $X \in \mathbb{R}^{n \times p}$, solve the normal equations $X^\top X \beta = X^\top y$ by Cholesky:

form $X^\top X$	np^2 (symmetric)
form $X^\top y$	$2np$
Cholesky of $X^\top X$	$\frac{1}{3}p^3$
two triangular solves	$2p^2$
$(X^\top X)^{-1}$ for standard errors	$\frac{4}{3}p^3$

In total $np^2 + \frac{1}{3}p^3$ flops ($np^2 + \frac{5}{3}p^3$ with standard errors): fastest when $n \gg p$, but forming $X^\top X$ squares the condition number (Lecture 4).

3.3 The sweep operator

For a symmetric $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ with $a_{kk} \neq 0$, **sweeping** on the kth diagonal entry maps A to $\hat{A} = (\hat{a}_{ij})$ with

$$\hat{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \hat{a}_{ik} = \frac{a_{ik}}{a_{kk}}, \qquad \hat{a}_{kj} = \frac{a_{kj}}{a_{kk}}, \qquad \hat{a}_{ij} = a_{ij} - \frac{a_{ik} a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

The last formula is one Gaussian elimination step (Schur complement update), so a sweep costs about n^2 flops ($n^2/2$ by symmetry), in place. The **inverse sweep** on entry k is

$$\check{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \check{a}_{ik} = -\frac{a_{ik}}{a_{kk}}, \qquad \check{a}_{kj} = -\frac{a_{kj}}{a_{kk}}, \qquad \check{a}_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}};$$

which restores A (e.g. $\hat{a}_{ij} - \hat{a}_{ik}\hat{a}_{kj}/\hat{a}_{kk} = a_{ij} - a_{ik}a_{kj}/a_{kk} + a_{ik}a_{kj}/a_{kk} = a_{ij}$).

Lemma 3.2 (Exchange property)

Let $U = [u_1, \ldots, u_n]$ and $V = [v_1, \ldots, v_n]$ be $p \times n$ with $V = UA$. After sweeping on entry k,

$$[v_1, \ldots, v_{k-1}, -u_k, v_{k+1}, \ldots, v_n] = [u_1, \ldots, u_{k-1}, v_k, u_{k+1}, \ldots, u_n]\,\hat{A}.$$

Sweeping exchanges u_k and v_k (up to sign).

Proof. Column k of the right side is $-v_k/a_{kk} + \sum_{i\neq k} u_i a_{ik}/a_{kk} = -(v_k - \sum_{i\neq k} u_i a_{ik})/a_{kk} = -u_k a_{kk}/a_{kk} = -u_k$, using $v_k = \sum_i u_i a_{ik}$. For $j \neq k$, column j is

$$v_k \frac{a_{kj}}{a_{kk}} + \sum_{i\neq k} u_i\Big(a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}}\Big) = \sum_{i\neq k} u_i a_{ij} + \frac{a_{kj}}{a_{kk}}\Big(v_k - \sum_{i\neq k} u_i a_{ik}\Big) = \sum_{i\neq k} u_i a_{ij} + a_{kj}u_k = v_j.$$

■

Proposition 3.3

Partition $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{k\times k}$ invertible. Sweeping on the diagonal entries $1, \ldots, k$ (in any order, as long as each pivot is nonzero) gives

$$\hat{A} = \begin{pmatrix} -A_{11}^{-1} & A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1} & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{pmatrix}.$$

Proof. Apply the lemma to $A = IA$ ($U = I$, $V = A$) for each sweep. After sweeping $1, \ldots, k$,

$$\begin{pmatrix} -I_k & A_{12} \\ 0 & A_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ A_{21} & I \end{pmatrix}\begin{pmatrix} \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{21} & \hat{A}_{22} \end{pmatrix}.$$

The blocks give $A_{11}\hat{A}_{11} = -I$, $A_{11}\hat{A}_{12} = A_{12}$, $A_{21}\hat{A}_{11} + \hat{A}_{21} = 0$, $A_{21}\hat{A}_{12} + \hat{A}_{22} = A_{22}$. The result depends only on the swept set, so sweeps commute. ■

Regression by sweeping. Sweeping $\left(\begin{smallmatrix} X^\top X & X^\top y \\ y^\top X & y^\top y \end{smallmatrix}\right)$ on its first p diagonal entries gives

$$\begin{pmatrix} -(X^\top X)^{-1} & (X^\top X)^{-1}X^\top y \\ y^\top X(X^\top X)^{-1} & y^\top y - y^\top X(X^\top X)^{-1}X^\top y \end{pmatrix} = \begin{pmatrix} -\sigma^{-2}\operatorname{Cov}(\hat{\beta}) & \hat{\beta} \\ \hat{\beta}^\top & \|y - \hat{y}\|^2 \end{pmatrix}:$$

coefficients, unscaled covariance and RSS at once. Hence **stepwise regression**: adding variable j is one sweep on entry j, removing it one inverse sweep, each $O(p^2)$; the swept matrix holds the current $\hat{\beta}$, standard errors and RSS, from which the F (or $t^2 = \hat{\beta}_j^2/\widehat{\operatorname{Var}}\hat{\beta}_j$) statistics are read off. A (numerically) zero pivot flags a variable collinear with those in the model. The price is the conditioning of the normal equations.

3.4 Selected exercises

Motivation. Cholesky versus LU with a determinant test, the entry bound behind symmetric pivoting, and one stepwise-regression sweep.

Exercise 3.1 ($LDL^\top$ and a determinant test)

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with nonsingular leading principal submatrices $A_k = A[1{:}k, 1{:}k]$, $k = 1, \dots, n$, and put $\det A_0 = 1$.
(a) Show that $A = LDL^\top$ for a unique unit lower triangular L and diagonal D, and that $d_k = \det A_k / \det A_{k-1}$.
(b) Show that A is positive definite if and only if $\det A_k > 0$ for all k, and that then $LD^{1/2}$ is the Cholesky factor of A.

Solution. [Supplement]

(a) *Existence.* By Theorem 2.2, $A = LU$ uniquely with L unit lower triangular, and $A_k = L_k U_k$ gives $\det A_k = \prod_{i\le k} u_{ii}$, so $u_{kk} = \det A_k / \det A_{k-1} \neq 0$. With $D = \operatorname{diag}(u_{11}, \dots, u_{nn})$ and $M = D^{-1}U$ unit upper triangular, $A = LDM$ and $A = A^\top = M^\top(DL^\top)$ is another LU decomposition; by uniqueness $M^\top = L$, i.e. $A = LDL^\top$.

Uniqueness. $A = L'D'L'^\top$ gives the LU decomposition $L'(D'L'^\top)$, so $L' = L$, $D'L^\top = DL^\top$ and $D' = D$.

(b) If A is positive definite, so is each A_k (x supported on the first k coordinates), and $\det A_k$, a product of positive eigenvalues, is positive. Conversely, if all $\det A_k > 0$ then all $d_k > 0$, and for $x \neq 0$, $y = L^\top x \neq 0$,

$$x^\top A x = y^\top D y = \sum_k d_k y_k^2 > 0.$$

So $A = (LD^{1/2})(LD^{1/2})^\top$ with positive diagonal $\sqrt{d_k}$, the Cholesky factor by uniqueness in Theorem 3.1. In Example 3.1 the pivots $4, \frac{1}{4}, \frac{9}{4}, \frac{1}{4}$ are the d_k, and $\ell_{kk} = \sqrt{d_k}$. ■

Exercise 3.2 (Entries of a positive semidefinite matrix)

Let A be symmetric positive semidefinite. Show:
(a) $|a_{ij}| \le \sqrt{a_{ii}a_{jj}}$ for all i, j;
(b) if $a_{ii} = 0$, then the whole i-th row and column of A vanish;
(c) $\max_{i,j}|a_{ij}| = \max_i a_{ii}$.

Solution. [Supplement] For $s, t \in \mathbb{R}$ and $i \neq j$,

$$0 \le (se_i + te_j)^\top A(se_i + te_j) = s^2 a_{ii} + 2st\, a_{ij} + t^2 a_{jj}. \qquad (\dagger)$$

(b) If $a_{ii} = 0$, $t = 1$ gives $2sa_{ij} + a_{jj} \ge 0$ for all s, so $a_{ij} = 0$ for $j \neq i$; by symmetry the column vanishes too.

(a) If $a_{ii} = 0$ or $a_{jj} = 0$, both sides are 0 by (b). If $a_{ii} > 0$, $s = -a_{ij}$, $t = a_{ii}$ in (†) give $0 \le a_{ii}(a_{ii}a_{jj} - a_{ij}^2)$. The case $i = j$ is trivial.

(c) By (a), $|a_{ij}| \le \sqrt{a_{ii}a_{jj}} \le \max\{a_{ii}, a_{jj}\}$, and the a_{ii} are among the $|a_{ij}|$.

Hence the largest remaining diagonal entry dominates the remaining block, justifying symmetric pivoting. ■

Exercise 3.3 (What one sweep does in stepwise regression)

Let $X_1 \in \mathbb{R}^{n\times q}$ and $x \in \mathbb{R}^n$ be such that $[X_1, x]$ has full column rank, $y \in \mathbb{R}^n$, $H_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$, $r_x = (I - H_1)x$ and $r_y = (I - H_1)y$. Order the cross-product matrix of

(X_1, x, y) as

$$T = \begin{pmatrix} X_1^\top X_1 & X_1^\top x & X_1^\top y \\ x^\top X_1 & x^\top x & x^\top y \\ y^\top X_1 & y^\top x & y^\top y \end{pmatrix}.$$

(a) Show that after sweeping on the first q diagonal entries, the lower-right 2×2 block is $\begin{pmatrix} \|r_x\|^2 & r_x^\top r_y \\ r_y^\top r_x & \|r_y\|^2 \end{pmatrix}$. In particular the next pivot is nonzero.
(b) Sweeping next on the entry of x, show that the residual sum of squares drops from $\|r_y\|^2$ to $\|r_y\|^2 - (r_x^\top r_y)^2/\|r_x\|^2$, and that the coefficient of x in the regression of y on $[X_1, x]$ is $r_x^\top r_y/\|r_x\|^2$.

Solution. [Supplement]

(a) Proposition 3.3 with $A_{11} = X_1^\top X_1$ (invertible), $A_{21} = [x, y]^\top X_1$, $A_{22} = [x, y]^\top [x, y]$ gives the swept block

$$A_{22} - A_{21} A_{11}^{-1} A_{12} = [x, y]^\top (I - H_1)[x, y].$$

As $I - H_1$ is symmetric idempotent, $u^\top (I - H_1) v = \{(I - H_1)u\}^\top (I - H_1)v$, giving the entries. $r_x \neq 0$ since $x \notin \mathcal{R}(X_1)$, so the pivot $\|r_x\|^2 > 0$.

(b) Sweeping on the x-entry with $a_{xx} = \|r_x\|^2$, $a_{xy} = r_x^\top r_y$, $a_{yy} = \|r_y\|^2$ gives

$$\hat{a}_{yy} = a_{yy} - \frac{a_{yx} a_{xy}}{a_{xx}} = \|r_y\|^2 - \frac{(r_x^\top r_y)^2}{\|r_x\|^2}, \qquad \hat{a}_{xy} = \frac{a_{xy}}{a_{xx}} = \frac{r_x^\top r_y}{\|r_x\|^2}.$$

By Proposition 3.3 this equals T swept on all entries of $X = [X_1, x]$, so by regression by sweeping the (y, y) entry is $\|y - \hat{y}\|^2$ and the (x, y) entry is the coefficient of x in $\hat{\beta} = (X^\top X)^{-1} X^\top y$. The drop is ≥ 0, zero iff $r_x \perp r_y$; the coefficient is the slope of r_y on r_x (Frisch–Waugh). ■